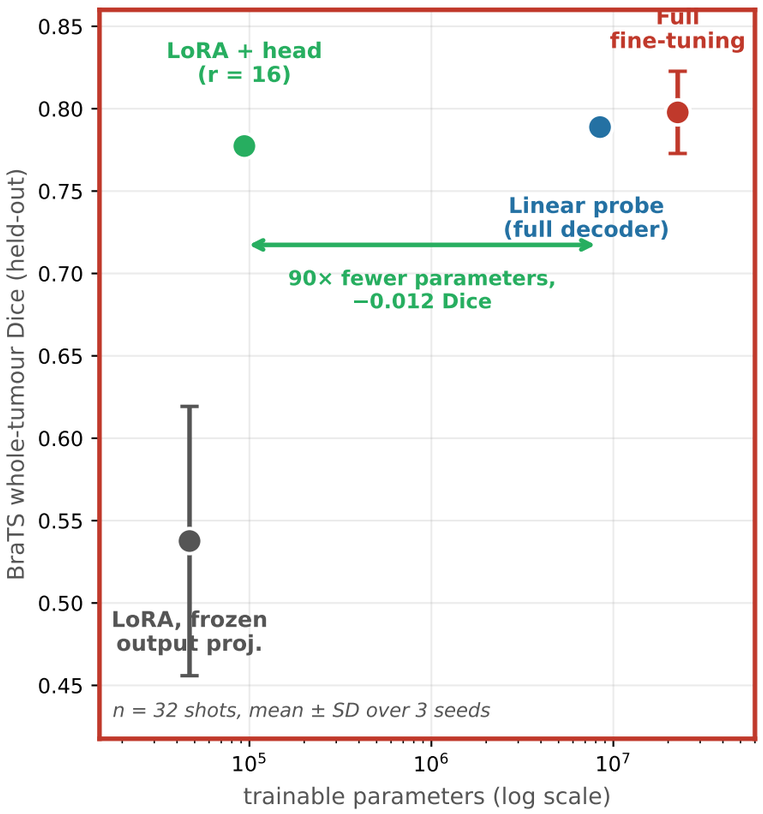
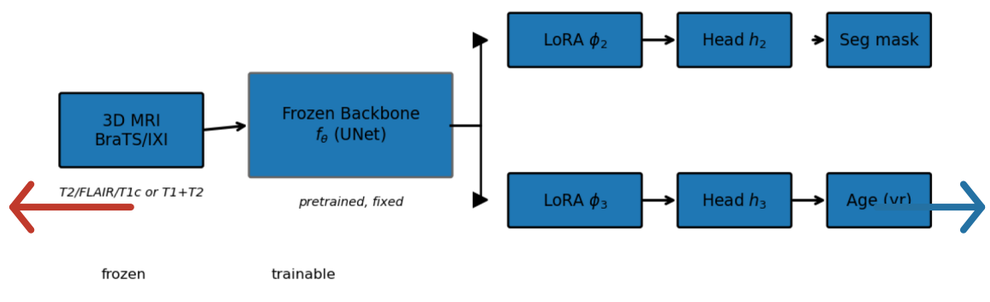
# Few-Shot Continual Learning for 3D Brain MRI with Frozen Foundation Models

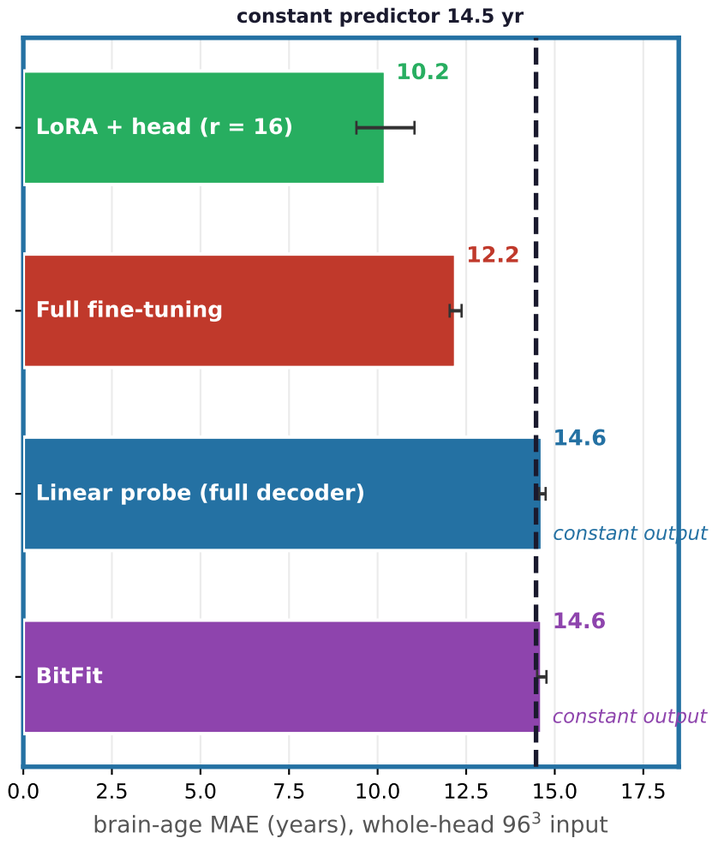
## Accuracy vs. trainable parameters




## Frozen backbone + per-task LoRA




## Brain age: only LoRA learns it




**10.2 yr MAE**

**Brain age: LoRA alone learns it**

vs. 14.5 yr constant baseline and
12.2 yr for full fine-tuning;
linear probing stays constant

**-0.012 Dice**

**Segmentation: 90× cheaper**

LoRA (r=16) + head: 0.777
vs. linear probing: 0.789
93.8K vs. 8.45M trainable

**66 params**

**The layer that matters**

Training the output projection
adds +0.22 Dice over LoRA
with it frozen (r = 8, paired)